

The Permutation Braid Image $B_3 \twoheadrightarrow S_3$

A non-scalar three-dimensional sigma-MGF test

Computational verification note

17 July 2026

Abstract

We construct the adjacent-transposition quotient of B_3 in the three-dimensional permutation representation. We derive its sigma-MGF coefficient formula from the three spectral types of S_3 , and independently recover the same integers as ranks of explicit Reynolds projectors on tensor products of symmetric powers.

1 Matrix group and braid relation

Let

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

These unitary matrices obey $ABA = BAB$. Sending $\sigma_1 \mapsto A$ and $\sigma_2 \mapsto B$ therefore defines a representation of B_3 . Matrix closure gives the six permutation matrices, so the image is S_3 acting on $V = \mathbb{C}^3$.

2 Proof and the three-class formula

For $W_\tau = \bigotimes_a \text{Sym}^{\tau_a} V$, expansion of each reciprocal determinant followed by Reynolds averaging proves

$$[m_\tau] \mathcal{S}_{S_3} = \dim(W_\tau^{S_3}) = \frac{1}{6} \sum_{g \in S_3} \prod_a h_{\tau_a}(g),$$

where $\sum_{r \geq 0} h_r(g) t^r = \det(I - tg)^{-1}$. The three conjugacy classes have eigenvalues

$$(1, 1, 1), \quad (1, 1, -1), \quad (1, \omega, \omega^2), \quad \omega^3 = 1.$$

Their generating series are respectively

$$\frac{1}{(1-t)^3}, \quad \frac{1}{(1-t)(1-t^2)}, \quad \frac{1}{1-t^3}.$$

Consequently, with

$$A_r = \binom{r+2}{2}, \quad B_r = \left\lfloor \frac{r}{2} \right\rfloor + 1, \quad C_r = \mathbf{1}_{3|r},$$

the proposed class formula becomes the completely explicit identity

$$\boxed{[m_\tau] \mathcal{S}_{S_3} = \frac{1}{6} \left(\prod_a A_{\tau_a} + 3 \prod_a B_{\tau_a} + 2 \prod_a C_{\tau_a} \right)}.$$

This is also the character-and-power-map formula: Newton's recurrence $rh_r = \sum_{k=1}^r \text{Tr}(g^k) h_{r-k}$ is obtained by taking the logarithm of the determinant generating series.

3 Direct versus formula verification

The direct route embeds the normalized occupation basis of $\text{Sym}^r V$ into $V^{\otimes r}$, restricts $g^{\otimes r}$ to that basis, forms the tensor product over the parts of τ , and ranks the averaged matrix. The two formula routes compute the power-trace recurrence over all six matrices and over the three constructed conjugacy classes. Thus the direct route does not diagonalize a Reynolds projector by using its character in advance.

τ	$\dim W_\tau$	projector rank	six elements	three classes
(1)	3	1	1	1
(2)	6	2	2	2
(1, 1)	9	2	2	2
(3)	10	3	3	3
(2, 1)	18	4	4	4
(2, 2)	36	8	8	8

All matrices close to exactly six elements, exactly three conjugacy classes are recovered, and the largest projector error is below 8×10^{-16} .

4 Reproduction

From the repository root run

```
.venv/bin/python for_this_guy/finite_braid_images/
s3_permutation/verification.py
.venv/bin/marimo edit for_this_guy/finite_braid_images/
s3_permutation/notebook.py
```

after joining wrapped lines. This example verifies the universal finite-image claim in a non-scalar braid quotient; it is not asserted to be a particular fusion-space representation of a chosen modular category.